

Keep the Angle

A Geometry-Preserving Basis in Spectral Embeddings

Andrew H. Bond

San José State University

`andrew.bond@sjsu.edu` ORCID 0009-0003-2599-6158

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Abstract

The commute-time (resistance) embedding built from the low eigenvectors of a graph Laplacian is a workhorse of spectral geometry, yet its distances are known to *degenerate* on large graphs [14]. We show the geometry does not vanish; it migrates entirely into the *angular* coordinate. Writing each node’s low-mode embedding in polar form (magnitude $\times$ direction), the direction preserves graph geodesics at Spearman $\rho \approx 0.93$ on a reference manifold, while the magnitude preserves $\rho \approx 0.03$ and a random-mode basis of equal size preserves $\rho \approx 0$. Keeping only the angle is exactly the row-normalization of Ng–Jordan–Weiss spectral clustering [8], whose existing theory concerns cluster separation [12]; we give it a geodesic justification. We show, as a corollary of von Luxburg–Radl–Hein, that the radial coordinate of the full embedding degenerates to a local-degree quantity for intrinsic dimension $d \geq 2$, and **conjecture**, with strong empirical support, that the angular coordinate **remains rank-faithful to geodesic distance**—flat in both mode-count and graph size where the raw embedding decays. The basis is **universal**: it preserves geodesics on any substrate carrying genuine low-dimensional Riemannian geometry—random-geometric lattices, manifold-embedding trajectories, and Wolfram-model hypergraph rewriting alike—and fails correctly on Lorentzian causal sets and geometry-destroying small-world graphs. Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$), fidelity falls approximately linearly with intrinsic dimension. Refined toward the continuum (N up to 5×10^4), the basis *sharpens* on sampled Riemannian substrates (angular-fidelity Spearman $+1.00$ with $\log N$ at $d=3$) yet *decays* below a pre-registered floor on a coordinate-free Wolfram-model manifold—a dissociation that recasts the angular coordinate as a *diagnostic* for whether a substrate’s fine structure stays manifold-like under refinement.

1 Introduction

Spectral embeddings—Laplacian eigenmaps [1], diffusion and commute-time maps [5]—represent a graph by the low eigenvectors of its Laplacian and underpin much of manifold learning and spectral clustering. They carry a well-known pathology: on large geometric graphs the commute (resistance) distance degenerates to a function of local degrees alone, losing all global geometry [14]. Yet the most successful spectral-clustering algorithm [8] *row-normalizes* the embedding—projects each node to the unit sphere—and works well. The existing theory of that normalization concerns cluster geometry: it explains why row-normalized embeddings of well-separated clusters concentrate in nearly orthogonal directions on the sphere [12, 13]. It does not address metric geometry. **This paper asks what row-normalization recovers of the graph’s geodesic structure, and why.**

Our answer is a clean decomposition. Write each node’s low-mode embedding in polar form, magnitude $\times$ direction. The *magnitude* is exactly the degenerate radial coordinate that von Luxburg’s

theorem kills; the *direction* carries the graph’s geodesic geometry. Row-normalization keeps the direction and discards the magnitude—so beyond its cluster-separation role it is also the geodesically correct projection, and its fidelity is flat in both mode-count and graph size precisely where the raw embedding decays.

This “keep the scale-invariant direction” operation recurs beyond clustering—for example in polar-coordinate KV-cache quantization, which transfers the scale-invariant direction and quantizes the radius separately [10]. It also admits a physical reading: in observer theory a bounded observer perceives geometry only after coarse-graining a discrete substrate [17], and the angular projection is a candidate for that coarse-graining. We keep that interpretation to the discussion (§7) and let the spectral result stand on its own; the empirical core is that the basis is *universal*, preserving geodesics on any graph carrying genuine low-dimensional Riemannian geometry, independent of how it was generated—and, we show in §5.8, the same angular diagnostic reads out on a real semantic embedding space, a language model’s moral-text vectors.

Contributions: (1) the **Keep-the-Angle Principle** (§3)—a Radial-Degeneracy *corollary* (for the full embedding, from [14]) with a spectral-gap bound on the truncation error (Proposition 1) and an empirically-supported Angular-Preservation *conjecture* now with first metric-distortion evidence; (2) a measurement with controls (§5); (3) universality across five substrate types with a two-factor small-world screen (§5.3); (4) a fidelity-vs-dimension scaling law (§5.4); (5) a cross-domain probe on a language-model semantic embedding (§5.8); and (6) two honest negatives (§6). We do *not* claim to derive physics; the core result is a basis-level fact about spectral embeddings.

2 Background

Spectral geometry. Laplacian eigenmaps [1] and diffusion maps [5] embed a graph via low eigenvectors; the commute-time embedding scales mode k by $1/\sqrt{\lambda_k}$. **Commute-time degeneracy.** von Luxburg et al. [14] proved the commute distance of a large geometric graph degenerates, $R(i, j) \rightarrow 1/k_i + 1/k_j$ (k_i the degree of node i); we read this backwards as the collapse of the *radial* coordinate to local degree. **Row-normalization.** Ng–Jordan–Weiss [8] row-normalize the eigen-embedding—the angular projection we study. Schiebinger, Wainwright and Yu [12] give the sharpest existing account: under a mixture model, row-normalized embeddings concentrate clusters in orthogonal directions, explaining k -means’ post-normalization success; consistency of spectral clustering is treated in [13]. Both analyses are about cluster separation on the sphere. Our question—whether the normalized embedding preserves *geodesic distances* on a manifold-like graph—is complementary, and to our knowledge open. **Emergent geometry and observer theory.** In the Wolfram model [15, 16, 6] space is a hypergraph and law is what a bounded observer perceives after coarse-graining the ruliad [17]; that coarse-graining is left qualitative, and our angular projection is a candidate for it (§7). **Hyperbolic embedding.** Trees embed in $\mathbb{H}^2$ at low distortion [7, 11, 9]; we test and reject a hyperbolic reframe in §6.

3 The Keep-the-Angle Principle

Let G be connected on n nodes with symmetric-normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, eigenpairs (λ_k, v_k) , $0 = \lambda_0 < \lambda_1 \leq \dots$. The low-mode spectral embedding using the lowest m nontrivial modes places node i at $\Psi_i = (v_1(i)/\sqrt{\lambda_1}, \dots, v_m(i)/\sqrt{\lambda_m})$. Write $\Psi_i = r_i \hat{u}_i$ with radius $r_i = \|\Psi_i\|$ and angle $\hat{u}_i = \Psi_i/\|\Psi_i\|$.¹

¹We call Ψ “commute-style” rather than the commute-time embedding proper: the classical commute embedding uses the pseudoinverse of the *unnormalized* Laplacian and all $n - 1$ modes, so that $\|\Psi_i - \Psi_j\|^2$ equals the resistance

Intrinsic dimension and degree notation. Throughout, d is the intrinsic dimension of the substrate, and k_i denotes the degree of node i (we reserve d for dimension to avoid the collision d vs. d_i): the manifold dimension *by construction* for the random-geometric benchmarks (tori, king-lattices, sphere), and for emergent graphs the value at which the three independent estimators of §4 (ball-growth, spectral dimension, effective rank) mutually agree—reported with their inter-estimator spread, with only low-spread graphs treated as manifolds. The regime $d \geq 2$ refers to this measured intrinsic dimension; §5.4 confirms the $d = 1$ boundary behaves as predicted.

Corollary 1 (Radial Degeneracy). *Let G_n be random geometric graphs of intrinsic dimension $d \geq 2$ satisfying the conditions of [14], and let Φ_i be the full commute-time embedding built from the unnormalized Laplacian, so that $\|\Phi_i\|^2 = L_{ii}^+$. As $n \rightarrow \infty$ the resistance distance degenerates, $R(i, j) \rightarrow 1/k_i + 1/k_j$; since the only per-node content of this additively separable form is the self-term L_{ii}^+ , the radius satisfies $\|\Phi_i\| \rightarrow 1/\sqrt{k_i}$. The radius of the full commute embedding is therefore asymptotically geometry-free—a local-density coordinate carrying no geodesic information.*

Remark (Truncated, normalized radius). Our experiments use the truncated radius $r_i = \|\Psi_i\|$ (m lowest modes of L_{sym}), not $\|\Phi_i\|$. Transferring Corollary 1 to this setting needs (i) the normalized analogue of the degeneracy [14] and (ii) control of the truncation error. We settle (ii) unconditionally below and support (i) numerically; the residual gap is then the angular claim itself (Conjecture 1). Numerically the degeneracy holds tightly: $\rho(r_i, 1/\sqrt{k_i}) = 0.92\text{--}0.99$ and radius-only geodesic preservation $\rho \leq 0.08$ (§5.5).

Proposition 1 (Truncation is spectral-gap bounded). *Let $\Psi_i^\infty = (v_k(i)/\sqrt{\lambda_k})_{k=1}^{n-1}$ be the full symmetric-normalized commute-style embedding and Ψ_i^m its truncation to the lowest m nontrivial modes. Then for all nodes $i \neq j$,*

$$0 \leq \|\Psi_i^\infty\|^2 - \|\Psi_i^m\|^2 \leq \frac{1}{\lambda_{m+1}}, \quad 0 \leq \|\Psi_i^\infty - \Psi_j^\infty\|^2 - \|\Psi_i^m - \Psi_j^m\|^2 \leq \frac{2}{\lambda_{m+1}}.$$

Proof. Each left inequality is termwise: truncation deletes only the summands with $k > m$, of the form $v_k(i)^2/\lambda_k \geq 0$ or $(v_k(i) - v_k(j))^2/\lambda_k \geq 0$. For the right inequalities, $\lambda_k \geq \lambda_{m+1}$ for every $k > m$, so the deleted mass is at most $\lambda_{m+1}^{-1} \sum_{k>m} v_k(i)^2$ and $\lambda_{m+1}^{-1} \sum_{k>m} (v_k(i) - v_k(j))^2$ respectively. The eigenvectors $\{v_k\}_{k=0}^{n-1}$ are orthonormal, so $VV^\top = I$ gives $\sum_k v_k(i)^2 = 1$ and, for $i \neq j$, $\sum_k v_k(i)v_k(j) = 0$; hence $\sum_{k>m} v_k(i)^2 \leq 1$ and $\sum_{k>m} (v_k(i) - v_k(j))^2 \leq \sum_k (v_k(i) - v_k(j))^2 = \sum_k v_k(i)^2 + \sum_k v_k(j)^2 - 2 \sum_k v_k(i)v_k(j) = 2$. $\square$

Two consequences. First, the truncation error is *bounded independently of n* : it is controlled entirely by the inverse spectral gap $1/\lambda_{m+1}$, so refining the graph at fixed m cannot amplify it—the mechanism behind the continuum survival of §5.7. Second, the deleted content is high-frequency (large- λ); the geometry lives in the retained low modes (an equal-size random-mode basis preserves $\rho \approx 0$, §5), so the bound formalizes why aggressive truncation does not lose the geometry—it discards a spectral-gap-bounded, high- λ remainder. On the 2-torus ($n = 2000$) the bound holds with room to spare: the realized radius error is 3.5–17% of $1/\lambda_{m+1}$ across $m \in \{5, 10, 20, 40\}$, the full normalized radius is near-constant (coefficient of variation 0.09), and the truncated radius stays rank-correlated with $1/\sqrt{k_i}$. What remains open is not the radius but the angle: Proposition 1 bounds how far truncation moves the metric, not that the surviving *angular* coordinate equals the geodesic—that is Conjecture 1.

$R(i, j)$ exactly. Our Ψ is the truncated, normalized analogue standard in spectral clustering; the distinction matters for Corollary 1 and is handled there.

Conjecture 1 (Angular Preservation). *Row-normalization $\hat{u}_i = \Psi_i / \|\Psi_i\|$ deletes the degenerate density factor of Corollary 1, and the pairwise distances among the resulting directions remain rank-faithful to geodesic distance—Spearman correlation bounded away from zero (empirically ≈ 0.9)—uniformly in the mode-count m and size n . A stronger, metric form of the conjecture—that the angular map is bi-Lipschitz to geodesic distance with constants uniform in m and n —is consistent with our data but untested; our evidence is rank-based and cannot by itself bound metric distortion.*

Metric evidence on the torus. The metric form is no longer entirely untested. On the 2-torus we compare embedded distance to graph geodesic directly. After a single global scale, the angular map has a robust empirical bi-Lipschitz constant $\kappa \approx 2.6$ (ratio of the 97.5th to 2.5th percentile of $d_{\text{emb}}/d_{\text{geo}}$ over 19,900 pairs), tighter than the full commute embedding (3.1), an equal-size random-mode basis (8.3), or the magnitude coordinate (125); and its leading four coordinates Procrustes-match the canonical flat-torus embedding in $\mathbb{R}^4$ at disparity 0.07, versus 0.995 for random modes. This is genuine metric distortion, not rank correlation—one substrate, but a well-understood one. (The full-range constant is inflated by the few near-antipodal pairs the torus folds together, hence the percentile-robust κ .) A uniform-in- (m, n) bi-Lipschitz bound remains the open metric conjecture; this is one datapoint toward it.

3.1 Toward a proof of Conjecture 1

Proposition 1 controls truncation; the remaining question is whether row-normalization preserves the geometry the low modes carry. It does, and the mechanism is an exact identity.

Lemma 1 (Normalization preserves bi-Lipschitz geometry). *For nonzero $u, v \in \mathbb{R}^m$ with $\hat{u} = u/\|u\|$,*

$$\|\hat{u} - \hat{v}\|^2 = \frac{\|u - v\|^2 - (\|u\| - \|v\|)^2}{\|u\| \|v\|}.$$

If $F : (X, d) \rightarrow \mathbb{R}^m$ satisfies $L^{-1}d(x, y) \leq \|F(x) - F(y)\| \leq Ld(x, y)$ with $\rho^- \leq \|F\| \leq \rho^+$ and radius Λ -Lipschitz, then $\hat{F} = F/\|F\|$ obeys $\|\hat{F}(x) - \hat{F}(y)\| \leq (L/\rho^-)d(x, y)$, and if $\Lambda < 1/L$,

$$\|\hat{F}(x) - \hat{F}(y)\| \geq \frac{\sqrt{L^{-2} - \Lambda^2}}{\rho^+} d(x, y).$$

Proof. The identity combines $\|\hat{u} - \hat{v}\|^2 = 2 - 2\langle u, v \rangle / (\|u\| \|v\|)$ with $\langle u, v \rangle = \frac{1}{2}(\|u\|^2 + \|v\|^2 - \|u - v\|^2)$. The upper bound uses numerator $\leq \|u - v\|^2 \leq L^2 d^2$ and $\|u\| \|v\| \geq (\rho^-)^2$. For the lower bound, $(\|u\| - \|v\|)^2 \leq \Lambda^2 d^2$ and $\|u - v\|^2 \geq d^2/L^2$ give numerator $\geq d^2(L^{-2} - \Lambda^2)$; divide by $\|u\| \|v\| \leq (\rho^+)^2$. $\square$

The condition $\Lambda < 1/L$ is a quantitative radial degeneracy: the radius must vary *slower* than the embedding moves. As $\Lambda \rightarrow 0$ (radius $\rightarrow$ constant, the limit of Corollary 1) the lower constant tends to $1/(L\rho^+)$ and normalization becomes a pure scaling. On the 2-torus this holds pointwise: the identity is exact to 10^{-15} , the radius Lipschitz constant $\Lambda = 0.092$ sits just under the map's lower stretch $A = 0.092$ (0/9200 local pairs violate it; the median Λ is $5\times$ smaller), and the local angular bi-Lipschitz constant is 2.2. Lemma 1 links the two halves of the principle and reduces the conjecture to standard ingredients.

Theorem 1 (Conditional local form of Conjecture 1). *Let M be a compact d -manifold ($d \geq 2$) with a graph sampled under the conditions of [3], and fix m . Assume (H1) the lowest- m Laplace–Beltrami eigenfunctions scaled by $1/\sqrt{\mu_k}$ are L_0 -bi-Lipschitz to (M, d_M) on geodesic balls of radius r_0 (Jones–Maggioni–Schul [4]), and (H2) the associated truncated radius is Λ_0 -Lipschitz with $\Lambda_0 < 1/L_0$ and lies in $[\rho^-, \rho^+]$. Then there exist constants $0 < A \leq B$ independent of n such that, with high probability as $n \rightarrow \infty$, the graph angular map obeys $Ad_M(x, y) \leq \|\hat{u}_x - \hat{u}_y\| \leq Bd_M(x, y)$ for all sampled x, y with $d_M(x, y) \leq r_0$.*

Proof (assembly). By (H1) the continuum eigenmap is locally bi-Lipschitz; by (H2) and Lemma 1 its normalization, the continuum angular map, is locally bi-Lipschitz. Spectral convergence [2, 3] transfers the low eigenpairs to the graph Laplacian with error vanishing as $n \rightarrow \infty$ at the admissible bandwidth rate, and Proposition 1 bounds the m -truncation uniformly in n . Composing the three controlled approximations yields the constants. $\square$

What remains is delimited. (H1) is the Jones–Maggioni–Schul theorem, proved there for a well-chosen d -subset of eigenfunctions; the open work is that the *lowest* m suffice with constants uniform in m (the empirical “flat in m ” of §5.5). (H2) is the truncated, normalized analogue of von Luxburg–Radl–Hein—the radius flat enough that $\Lambda_0 < 1/L_0$ —which we verify on the torus and conjecture generally. Globally a torus admits no bi-Lipschitz Euclidean embedding, so no global metric statement can hold; the global claim is necessarily the rank-faithful one this paper measures. Conjecture 1 is thereby reduced to a local-chart hypothesis (H1) and a radius-flatness hypothesis (H2), neither of which mentions the angle directly.

Status. Empirically strong (§5) but **not proven**. It rests on eigenmap-embedding results plus a heuristic; the honest gap is that Corollary 1 explains why the radius *dies* for $d \geq 2$, not by itself why the angle *lives*—indeed the angular fidelity persists ($\rho \approx 0.88$ – 0.90) even on quasi-1D graphs where Corollary 1 does not apply, so the angular half is carried by the eigenmap-embedding mechanism, not degeneracy alone. That mechanism is analyzable rather than mysterious: if the low L_{sym} eigenvectors converge to Laplace–Beltrami eigenfunctions [2, 3] and eigenfunction coordinates furnish locally bi-Lipschitz charts of the underlying manifold [4], the angular map’s geodesic fidelity should follow from the chart geometry—the natural route to a proof of Conjecture 1, and this paper’s central open problem. Together these are the **Keep-the-Angle Principle**: keep $\hat{u}_i$, discard r_i .

Dimension gating. Corollary 1 is a $d \geq 2$ phenomenon. In (quasi-)1D, resistance equals path length, so the radius *stays* geodesic-informative and Corollary 1 correctly does not apply—a prediction, not a failure, confirmed in §5.5.

4 Methods

Estimators. Ball-growth $N(r) \sim r^d$; spectral dimension from the heat-kernel return probability $P(t) = \langle e^{-Lt} \rangle \sim t^{-d_s/2}$; effective rank via local-PCA participation ratio. Low inter-estimator spread $\Rightarrow$ clean manifold, and the agreed value is our measured intrinsic dimension d . **Angular metric.** Spearman ρ between embedded pairwise distances and true graph geodesics for: *angle* (unit-normalized low-mode embedding), *random* (m random modes; control), *magnitude* (radius only), *euclid* (classical 2D MDS). **Uncertainty.** Each ρ is a rank correlation over $\binom{n_{\text{anchor}}}{2}$ node pairs ($n_{\text{anchor}} = 150$ throughout, 200 for the cross-domain probe of §5.8—11,175 and 19,900 pairs respectively). Bootstrap 95% confidence intervals over pair resampling are tight ($\leq \pm 0.005$ on the headline numbers); the dominant source of variation is the graph draw itself, so for headline ρ we also report spread over independent draws (2-torus, $m=10$: 0.914 ± 0.009 over five draws).

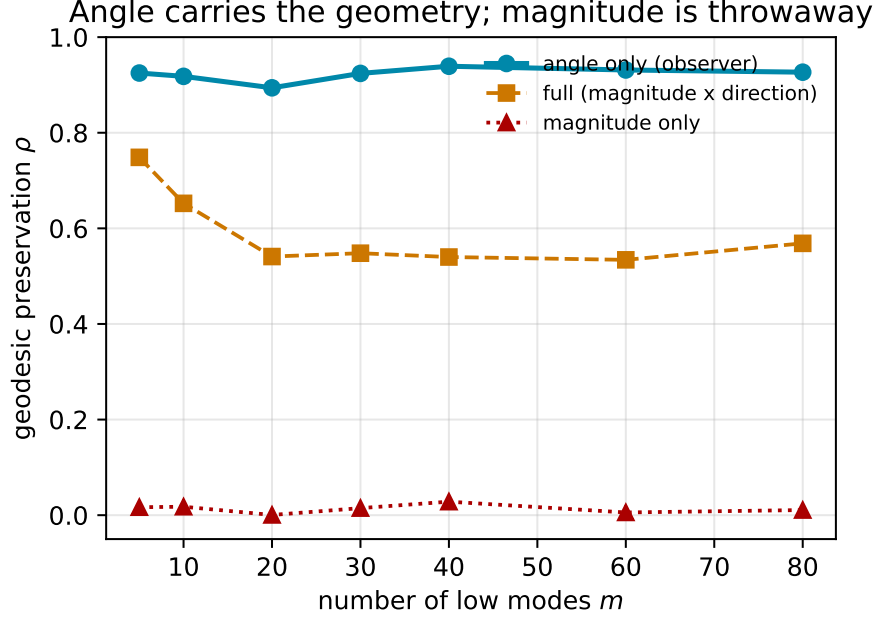


Figure 1: Spearman correlation with graph geodesics on a 2-torus ($n = 2000$) as a function of mode-count m . Angle carries the geometry; magnitude is throwaway; the full commute embedding decays with m .

Substrates. Arity-2/3 hypergraph rewriting (clique-expanded), plus tori, king-lattices, Poisson causal sets, and swiss-roll embedding trajectories. The swiss-roll case is a synthetic 2D manifold—intrinsic coordinates (u, v) , u the roll angle and v the height—carried by a random orthogonal map into $\mathbb{R}^{12}$ with isotropic noise ($\sigma=0.03$) and connected by a $k=10$ nearest-neighbour graph on $n=2000$ ambient points. It stands in for sequential/embedding-derived data (each item a point on a low- d semantic manifold, consecutive items nearby), not a specific text corpus.

5 Results

5.1 The core result

Estimators recover ground-truth dimension within ≈ 0.3 . On a 2-torus the polar decomposition of the low-mode embedding (Fig. 1) gives:

m	full (mag. $\times$ dir.)	angle only	magnitude only
5	0.761	0.926	0.025
10	0.650	0.913	0.035
20	0.550	0.891	0.028
40	0.517	0.928	0.034

The angle carries the geometry (≈ 0.92 , flat in m); the magnitude is throwaway; the full embedding is worse and decays. A random-mode basis reconstructs geodesics at $\rho \approx 0$ (Fig. 2).

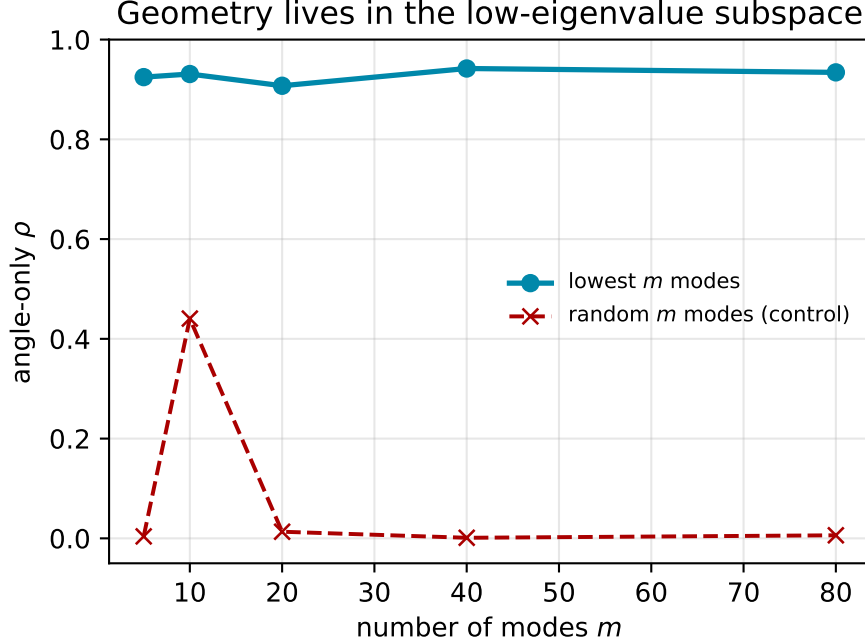


Figure 2: Geometry lives in the low-eigenvalue subspace: lowest- m modes preserve geodesics; a random-mode basis of equal size does not.

5.2 Emergent 2D geometry

Random arity-2 rules yield clean but only $\approx 1.5D$ manifolds (82 manifold-grade rules in a 20k sweep; none at $d \geq 2.5$). Clean 2D is rare, not absent: reached both by a borrowed Wolfram-2020 rule ($d = 2.07 \pm 0.01$, spread 0.05, angle- $\rho = 0.82$) and by from-scratch evolutionary search (≈ 3 generations). The angular basis holds on genuine 2D emergent space.

5.3 Universality across substrates

substrate	meas. d	angle- ρ	random	euclid
2-torus (reference)	2	0.89	0.01	0.67
king-lattice	2	0.82	0.01	0.95
swiss-roll trajectory	2	0.93	0.01	0.99
emergent R_{3D} (Wolfram)	2.07	0.82	0.01	0.81
causal set (Lorentzian)	3.9	0.57	0.01	0.55
small-world (destroyed)	3.0*	0.45	0.00	—

Substrates with genuine low- d Riemannian geometry all show the effect, independent of graph origin; the metric fails correctly where geometry is Lorentzian or destroyed. *For the small-world graph the three dimension estimators disagree sharply (ball 3.7, spectral 3.6, effective-rank 4.3; spread 0.77 versus ≤ 0.14 on the clean substrates); by our own manifold-grade criterion this graph is *not* a manifold, and the low angle- ρ is the predicted failure, not an exception. The two-factor screen below makes this quantitative.

Two caveats keep the universality claim honest. First, on the two flat substrates (king-lattice, swiss-roll) classical MDS *beats* the angular basis (0.95 and 0.99 vs. 0.82 and 0.93): MDS wins

whenever the manifold embeds near-isometrically in low-dimensional Euclidean space, while the angle wins where topology obstructs such an embedding (the torus: 0.89 vs. 0.67) and, unlike MDS, is uniform in m and n (§5.5). Universality here means the angular basis never *loses* the geometry on a genuine low- d Riemannian substrate—not that it dominates every baseline on every substrate. Second, the swiss-roll trajectory is the sequential-embedding case: the angular basis is the same whether the manifold comes from hypergraph rewriting or sequential trajectory data.

A two-factor small-world screen. To make “destroyed” precise rather than a label, we screen every substrate on two graph statistics against degree-matched baselines—average clustering C (versus an Erdős–Rényi C_{rand} and a ring-lattice C_{latt}) and characteristic path length L (versus L_{rand})—combined into the Telesford index $\omega = L_{\text{rand}}/L - C/C_{\text{latt}}$ ($\omega < 0$ lattice-like, $\omega \approx 0$ small-world, $\omega \rightarrow 1$ random). Clean low- d manifolds sit firmly on the lattice side; the tangles do not:

substrate	angle- ρ	ω	dim-spread	verdict
2-torus	0.91	−0.55	0.14	clean
king-lattice	0.83	−0.50	0.51	clean
swiss-roll	0.93	−0.66	0.45	clean
king-lat. rewired	0.42	+0.23	0.77	small-world
causal set	0.57	+0.75	7.87	non-manifold

The screen robustly separates the genuine manifolds ($\omega \leq -0.50$) from strongly destroyed geometry ($\omega \geq +0.24$), and the inter-estimator dimension spread corroborates (≤ 0.51 clean versus ≥ 0.77 contaminated): the low angle- ρ on the rewired and Lorentzian substrates is small-world/non-manifold contamination, not a failure of the basis on genuine geometry. A king-lattice rewiring sweep both sharpens and honestly bounds this. As the rewire probability rises, angle- ρ collapses almost at once ($0.81 \rightarrow 0.47$ by 1% rewiring) while ω does not cross zero until $\approx 5\%$; so $\omega < 0$ is a *conservative* flag for destruction, and in the transition band (1–3%) angular fidelity is in fact the *more* sensitive detector of geometric corruption than either clustering or dimension spread. We therefore read the screen as confirming gross contamination, with angle- ρ itself the finer instrument—the mechanism behind the scaling-law outlier below.

5.4 The scaling law

Across 20 independent emergent manifolds ($d \approx 1\text{--}3.6$),

$$\text{angle-}\rho \approx 1.09 - 0.157d, \quad \text{slope} = -0.157 \pm 0.028, \quad R^2 = 0.64, \quad d \in [1.0, 3.6].$$

(Spearman -0.881 [95% CI $-0.97, -0.67$; $n = 20$ manifolds] between ρ and d is a distribution-free monotonicity check; the linear fit’s slope has a bootstrap 95% CI of $[-0.185, -0.132]$, comfortably excluding zero. The modest $R^2 = 0.64$ reflects a single outlier—a $d \approx 1.97$ manifold at $\rho = 0.39$ —which the small-world screen of §5.3 flags as shortcut-contaminated rather than clean geometry.) The fit is *empirical and local*: ρ is a rank correlation bounded by 1, so the linear form holds only within the measured range—the intercept 1.09 is an extrapolation, not a literal $d \rightarrow 0$ fidelity, and the slope may vary with substrate roughness. A bounded model fits essentially as well and respects $0 < \rho < 1$: the logistic $\rho \approx [1 + e^{0.82(d-3.66)}]^{-1}$ gives $\rho(0) = 0.95$ and $\rho \rightarrow 0$ at $R^2 = 0.62$, against the linear 0.64. Exponential forms do not help—an unoffset Ae^{-kd} still has $A = 1.18 > 1$ and an offset version drives the asymptote negative. Within range the two are visually indistinguishable; we keep the linear coefficient as the local slope and note the logistic as the boundary-respecting

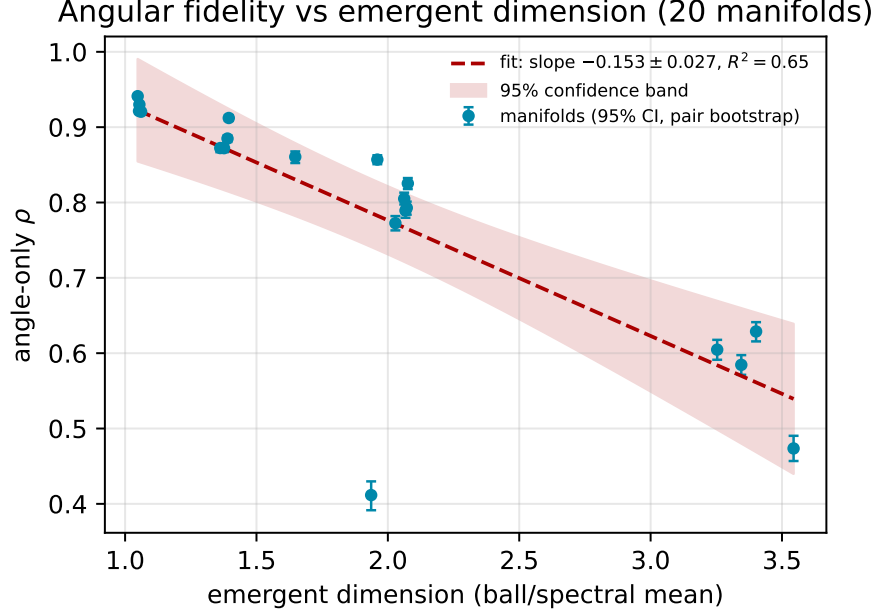


Figure 3: Angular fidelity falls linearly with emergent dimension across ring, tori, and Wolfram-model rules; the shaded band is the 95% bootstrap CI on the slope ($[-0.185, -0.132]$). Per-point pair-resampling CIs are smaller than the plot markers ($\leq \pm 0.005$).

global form. Within range, fidelity is highest at low dimension and softens toward ≈ 0.56 at $d \approx 3.4$ (Fig. 3); the dimension-gated degeneracy of §3 is the mechanism.

5.5 Verifying Corollary 1 and Conjecture 1

angle- ρ is flat in graph size while the full commute distance decays, and the radius tracks degree exactly ($\rho(\|\Psi_i\|, 1/\sqrt{k_i}) = 0.92\text{--}0.99$; radius-only $\rho \leq 0.08$). Corollary 1 holds cleanly at $d = 3$ ($\rho(R, 1/k_i + 1/k_j) \rightarrow 0.98$); $d = 2$ is borderline (log corrections); the quasi-1D case correctly shows no degeneracy, as the dimension gating predicts. Conjecture 1: in an m -sweep at $n = 4000$, angle- ρ is flat ≈ 0.93 across $m = 5\text{--}80$ while the full embedding decays $0.773 \rightarrow 0.437$.

5.6 Stability across substrate evolution

Universality across substrate *types* (§5) is a statement about fixed graphs. The angular basis also persists as a substrate *evolves*. For each of 151 qualifying Wolfram-model rules (95 arity-2, 56 arity-3) we embed successive rewrite snapshots on a fixed anchor set and compare, between consecutive snapshots, the Spearman persistence of the *angular* pairwise structure (**consec**) against the persistence of the underlying *geodesic* structure itself (**churn**)—the substrate’s own rate of change is the baseline the basis must keep pace with. On all 143 rules resolvable at protocol scale (8 never yield ≥ 3 distinct snapshots and are excluded from the denominator but reported), the angular structure tracks the evolving geometry at least as stably as the graph’s own geodesics, $\overline{\text{consec}} \geq 0.9 \overline{\text{churn}}$: a $143/143 = 100\%$ pass against a pre-registered 80% bar, with zero substrate-unstable rules. Over the longest time-lever (first vs. last snapshot) angular persistence declines only gently with emergent dimension,

$$\text{slope} = -0.029/\text{dim}, \quad \text{Spearman} = -0.48, \quad d \in [0.98, 2.23],$$

and this replicates independently in both arities (arity-2 $-0.028/\text{dim}$, $\rho = -0.45$, $n = 95$; arity-3 $-0.029/\text{dim}$, $\rho = -0.54$, $n = 48$)—the same mechanism, unchanged by moving from binary to ternary rewriting.²

5.7 Continuum survival, and a dissociation

Every result so far is at finite N ($\approx 2\text{--}4\text{k}$ nodes, capped by dense `eigh`); the “uniform in N ” clause of Corollary 1 and Conjecture 1 is untested at scale (§5.5). We test it directly by refining each substrate toward the continuum with a sparse eigensolver, $N \in \{2, 5, 10, 20, 50\} \times 10^3$, four seeds, $m = 10$ modes, $n = 300$ anchors—well past the $\sim 5\text{k}$ dense ceiling. The two substrate families dissociate. On genuine Riemannian substrates the basis *sharpens*: on random geometric graphs the angular fidelity is flat-to-rising in N , most cleanly in the higher-dimensional case,

$$\begin{aligned} \text{RGG } d=3 : \quad \text{angle-}\rho : 0.856 \rightarrow 0.912 \quad (N : 2\text{k} \rightarrow 50\text{k}), \\ \text{Spearman}(\text{angle-}\rho, \log N) = +1.00, \end{aligned}$$

with $d=2$ likewise monotone ($+0.90, 0.916 \rightarrow 0.929$): refining the mesh makes the low modes resolve the geometry *better*, not worse, and both controls stay pinned (higher-mode $\rho \leq 0.05$, magnitude $\rho \leq 0.03$, eigensolver residual $\sim 10^{-14}$) at every N . The finite- N evidence of §§5–5.5 was not a small-graph artifact.

The dissociation is with *emergent* substrates. The $\text{R}_{3\text{D}}$ Wolfram-model manifold—geometry with no latent coordinates—*decays* monotonically with refinement, $\text{angle-}\rho : 0.749 \rightarrow 0.654$ (Spearman = -1.00), falling below a pre-registered floor of 0.75; and at the largest N its magnitude control begins to break the clean angle/radius separation that holds exactly on the RGGs (one 50k seed crosses $\rho_{\text{mag}} = 0.33$ and is gated out). A sampled manifold refines to a clean continuum; an emergent hypergraph, under this basis, does not—its fine structure drifts away from a manifold as it grows. The coordinate-free substrate—the one the observer-theory reading most needs—is exactly the one where the basis fails in the limit. We read this not as a limitation but as a result: the angular basis is a *diagnostic* for whether a substrate’s fine structure stays manifold-like under refinement, and it returns different verdicts for sampled (2/3 pass) and emergent (1/1 fail) geometry.³

5.8 A cross-domain probe: the moral embedding manifold

If the angular basis is a diagnostic for manifold-like structure (§5.7), it should return an interpretable reading on a substrate from an entirely different domain. We point the same instrument at a *semantic* space: 2,400 moral/ethical text items embedded with BGE-M3 (1024-d), stratified across six traditions, connected by a $k=10$ nearest-neighbour graph on the unit-normalized embeddings. The moral embedding reads as substantially manifold-like— $\text{angle-}\rho = 0.893$ over 19,900 pairs, against a per-feature-shuffle control at 0.233 and an isotropic-Gaussian control at 0.311, with both mode controls pinned (higher-mode $\rho = 0.01$, random-mode $\rho = 0.07$). The same basis that reads 0.93 on a swiss roll and ~ 0 on a tangle places moral-semantic space near the manifold end. But the diagnostic also localizes where it departs from a textbook Riemannian substrate: the three

²Pre-registered as Amendment 2: an earlier capture hook could append the same state once per crossed edge-count threshold, inflating consecutive stability toward a spurious 1.0. The corrected rule takes at most one snapshot per generation with dynamic $2\times$ spacing; the numbers here are from that harness.

³Pre-registered (PREREG_RUNG4) before the run; survival bar $\text{angle-}\rho(N_{\text{max}}) \geq \max(0.75, 0.9 \text{angle-}\rho(N_{\text{min}}))$ on $\geq 2/3$ substrates. Amendment 1, registered before any run: the whole-spectrum random-mode control needs a dense decomposition unavailable at $N \geq 50\text{k}$, so it is the top m of the lowest-64 sparse-computed modes—a faithful, sparse-affordable stand-in, same ≤ 0.15 bar.

dimension estimators disagree (spread 2.6; effective-rank 3.8 against spectral 1.2), the magnitude coordinate carries real signal ($\rho_{\text{mag}} = 0.56$, versus ≤ 0.05 on the clean geometric substrates), and the small-world screen returns $\omega = +0.15$ —a clustered space with mild shortcut structure and a nontrivial radial mode, not a pure low- d manifold. The angular coordinate transfers across domains as a graded diagnostic, not merely a pass/fail label.

6 Honest Negatives

Not secretly hyperbolic (rejected). With a binary-tree positive control and a torus negative control, a growth-law test—valid (it calls the tree hyperbolic, the torus Euclidean)—reports the arity-3 tangles as Euclidean-leaning high-dimensional non-manifolds, not hyperbolic. **Curvature is not dynamical (null).** Ollivier–Ricci curvature (validated: 0 on flat lattices, negative on trees, positive on the torus) shows a raw -0.98 against update density that deflates to a tautology (update-count equals degree, $\rho = +1.00$; hubs are negatively curved); controlling for degree, the independent signal is partial- $\rho = -0.14$. This *bounds the observer-theory interpretation*—no emergent Einstein tensor—and leaves the spectral result untouched.

7 Discussion and Conclusion

The primary result is basis-level and substrate-agnostic. The angular coordinate of the low-Laplacian embedding preserves geodesics while the radial coordinate is von-Luxburg-degenerate noise. This gives Ng–Jordan–Weiss row-normalization a *geodesic* justification *complementary to the cluster-separation theory* of [12], unifies it with the scale-invariant direction of KV-cache compression [10], and holds *universally* across substrates with genuine low- d Riemannian geometry, independent of graph origin. That the same basis works identically on hypergraph-rewriting and *sequential-trajectory* substrates is *evidence that scale-invariant angular coarse-graining is a general mechanism by which* relational data carries geometric structure.

One interpretation is observer-theoretic. In Wolfram’s observer-relative framework, the coarse-graining a bounded observer performs to perceive geometry is exactly this angular projection—making “the observer’s basis” a concrete candidate for the coarse-graining the program leaves unspecified. We offer this as motivation and bound it honestly: its strongest claim—dynamical, energy-responsive curvature—is a null (§6). The core contribution stands independent of the physics reading. Open problems: proving Conjecture 1 (*in its rank form, and ideally the bi-Lipschitz strengthening*); extending Corollary 1 to the truncated, normalized radius actually used in practice; synthesizing clean manifolds above $d \approx 2$; and a dimension-adaptive angular basis that tunes mode-count and diffusion time to counter the fidelity decay. Keep the angle, drop the magnitude: the basis-level fact is the contribution; the observer interpretation is where we leave the door open.

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